



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2023/2024

Course Instance Codes	3BA1, 3BME1, 4BMU1, 1OA2, 4BCT1, 3BMS2, 3BS9, 3FM2, 4BS2
Exams	3rd and 4th year (Arts and Science), Study Abroad
Module Code	MA342
Module	Topology
External Examiner	Prof. Colva Roney-Dougal
Internal Examiner	Dr Niall Madden *Dr Tobias Rossmann
<u>Instructions:</u>	Answer all four questions. Each question carries 25 marks. All workings must be shown.
Duration	2 hours
No. of Pages	3 pages, including this one
Discipline	Mathematics
<u>Requirements:</u>	No special requirements
Release to Library	Yes
Release in Exam Venue	Yes
MCQ	No
Statistical / Log Tables	No
Non-programmable Calculator	Yes
Mathematical Tables	No
Graph paper	No

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- Q1.** (a) Let X be a set.
- (i) Define what is meant by a *topology* on X .
 - (ii) Give an example of a finite set X and two distinct topologies on X . [5 marks]
- (b) (i) Let X be a topological space and let $A \subseteq X$ be a subset. Define the *subspace topology* on A .
- (ii) Let $\mathbf{R}$ denote the set of real numbers endowed with the standard topology. Show that the subspace topology on the rational numbers $\mathbf{Q}$ is not discrete. [5 marks]
- (c) Define what is meant by a *base* for a topology. Describe a base for the *lower limit topology* on the set of real numbers $\mathbf{R}$. [5 marks]
- (d) Let X be a topological space. Let $A \subseteq X$.
- (i) Define the *closure* $\bar{A}$ of A .
 - (ii) Show that if $A \cap N \neq \emptyset$ for each neighbourhood N of a point $x \in X$, then x belongs to $\bar{A}$. [5 marks]
- (e) Determine (with justification) the closure of $(0, 1)$ in the lower limit topology on $\mathbf{R}$. [5 marks]
- Q2.** (a) Define what it means for a topological space to be *compact*. [4 marks]
- (b) Is $\mathbf{R}$ endowed with the standard topology compact? Justify your answer. [5 marks]
- (c) Briefly explain why any two of the following topological spaces (with their usual topologies) are non-homeomorphic. There are six comparisons.
- (i) $[0, 1]$
 - (ii) $(0, 1)$
 - (iii) $\mathbf{S}^1$ (the unit circle)
 - (iv) $\mathbf{R} \setminus \{0\}$ [12 marks]
- (d) Let X and Y be topological spaces. Write down a base for the *product topology* on $X \times Y$. [4 marks]

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- Q3.** (a) Let X denote the set of real numbers endowed with the cofinite topology. Is X compact? Is X Hausdorff? Justify your answers. [6 marks]
- (b) (i) Let $f: X \rightarrow Y$ be a continuous surjection between topological spaces. Show that if X is connected, then so is Y .
- (ii) Briefly explain how the preceding part implies the Intermediate Value Theorem from real analysis. [8 marks]
- (c) Let $f: X \rightarrow Y$ be a continuous bijection between compact Hausdorff spaces. Show that f is a homeomorphism. [5 marks]
- (d) Briefly show that the unit interval $[0, 1]$ is a continuous image of the Cantor set C . (You do not need to prove continuity or surjectivity of the function that you define.) Conversely, is C a continuous image of $[0, 1]$? Justify your answer. [6 marks]
- Q4.** (a) Let X be a topological space and let $\sim$ be an equivalence relation on X . Define the *quotient topology* on $X/\sim$. [4 marks]
- (b) Show that $\mathbf{S}^1$ is a quotient of $[0, 1]$. [5 marks]
- (c) Let X and Y be topological spaces.
- (i) Let $f: X \rightarrow Y$ and $g: X \rightarrow Y$ be continuous functions. Define what it means for f and g to be *homotopic*.
- (ii) Define what it means for X and Y to be *homotopy equivalent*. [6 marks]
- (d) Define what it means for a topological space to be *simply connected*. [4 marks]
- (e) Show that every star-shaped subset $S \subseteq \mathbf{R}^n$ is simply connected. [6 marks]